

Belt Friction Practice Problems

3. The coefficient of friction is $\mu = 0.2$ between the fixed posts at B and C and the cord. Determine the minimum force P required to initiate movement of the weight D upward.

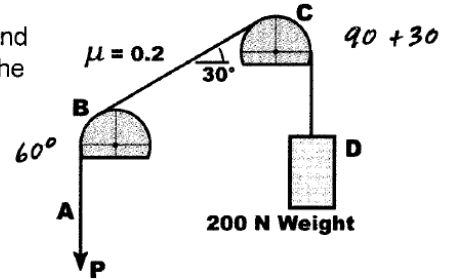
- A. 375 N
- B. 152 N
- C. 103 N
- D. 432 N
- E. 573 N

$$\theta = 180 = \pi$$

$$T_2 = T_1 e^{\mu \pi}$$

$$= 200 (1.874)$$

$$T_2 = 374.9 \text{ N}$$



17. (5 pt) A cable ABCD passes over two fixed posts B and C. A 70 N weight is suspended from C and a 20 N weight is suspended from B. Determine the coefficient of friction (same for both posts) μ at which the 70 N weight is on the verge of moving down and the 20 N weight is on the verge of moving up.

- A. 0.45
- B. 0.3
- C. 0.35
- ☒ D. 0.4
- E. 0.5

$$\mu = .4$$

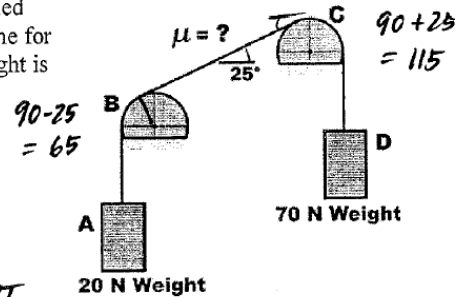
$$\theta = 115 + 65$$

$$= 180 = \pi$$

$$70 = 20 \cdot e^{\mu \pi}$$

$$\frac{70}{20} = 3.5 = e^{\mu \pi}$$

$$\ln 3.5 = 1.253 = \mu \pi$$



16. The coefficient of friction is $\mu = 0.15$ between the fixed posts at B and C and the cord. What is the mass m of D required so that motion of D impends downward?

- A. 60 kg
- B. 70 kg
- ☒ C. 80 kg
- D. 90 kg
- E. 100 kg

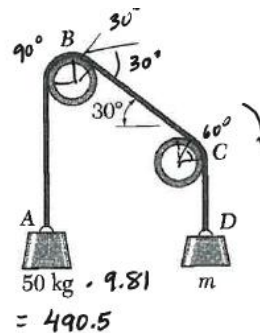
$$\theta = 90 + 30 + 60$$

$$= 180 \cdot \frac{\pi}{180} = \pi \text{ rad}$$

$$T_2 = T_1 e^{\mu \theta}$$

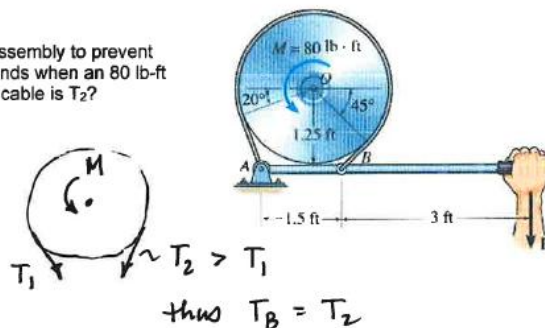
$$= (490.5) e^{(0.15)(\pi)}$$

$$= \frac{785.8 \text{ N}}{9.81} = 80.0 \text{ kg}$$



17. Force P is applied to the handle and belt assembly to prevent motion of the drum. If motion of the drum impends when an 80 lb-ft torque is applied to the drum as shown, which cable is T_2 ?

- A. Cable A
- ☒ B. Cable B



4. A cable is used to prevent a 100 lb weight **W** from falling. The cable is wrapped two full revolutions around a fixed post. The coefficient of friction is $\mu = 0.2$ between the post and the cable. Determine the minimum force **T** which must be applied to the upper end of the cable to prevent the weight from moving downward.

- A. 8.1 lb ✓
B. 52 lb
C. 1235 lb
D. 25.8 lb
E. 77.4 lb

$$\beta = 2 \text{ rev} = 4\pi$$
$$T_2 = 100 = T_1 e^{-2(4\pi)}$$
$$= T_1 \cdot 12.35$$

$$T_1 = 8.1 \text{ lb}$$

